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Appendix

A Mathematical details of CCA and SVCCA

Canonical Correlation of X, Y Finding maximal correlations between X, Y can be expressed as finding a, b to maximise:

$$\frac{a^T \Sigma_{XY} b}{\sqrt{a^T \Sigma_{XX} a}\sqrt{b^T \Sigma_{YY} b}}$$

where $\Sigma_{XX}, \Sigma_{XY}, \Sigma_{YX}, \Sigma_{YY}$ are the covariance and cross-covariance terms. By performing the change of basis $\tilde{\boldsymbol{x}}_1 = \Sigma_{xx}^{1/2} a$ and $\tilde{\boldsymbol{y}}_1 = \Sigma_{YY}^{1/2} b$ and using Cauchy-Schwarz we recover an eigenvalue problem:

$$\tilde{\boldsymbol{x}}_1 = \text{argmax}\left[\frac{x^T \Sigma_{XX}^{-1/2} \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX} \Sigma_{XX}^{-1/2} x}{||x||}\right] \tag{*}$$

SVCCA Given two subspaces $X = \{\boldsymbol{x}_1, ..., \boldsymbol{x}_{m_1}\}, Y = \{\boldsymbol{y}_1, ..., \boldsymbol{y}_{m_2}\}$, SVCCA first performs a singular value decomposition on X, Y. This results in singular vectors $\{\boldsymbol{x'}_1, ..., \boldsymbol{x'}_{m_1}\}$ with associated singular values $\{\lambda_1, ..., \lambda_{m_1}\}$ (for X, and similarly for Y). Of these m_1 singular vectors, we keep the top m_1' where m_1' is the smallest value that $\sum_{i=1}^{m_1'} |\lambda_i| (\geq 0.99 \sum_{i=1}^{m_1} |\lambda_i|)$. That is, 99% of the variation of X is explainable by the top m_1' vectors. This helps remove directions/neurons that are constant zero, or noise with small magnitude.

Then, we apply Canonical Correlation Analysis (CCA) to the sets $\{\boldsymbol{x'}_1, ..., \boldsymbol{x'}_{m_1'}\}, \{\boldsymbol{y'}_1, ..., \boldsymbol{y'}_{m_2'}\}$ of top singular vectors.

CCA is a well established statistical method for understanding the similarity of two different sets of random variables – given our two sets of vectors $\{\boldsymbol{x'}_1, ..., \boldsymbol{x'}_{m_1'}\}, \{\boldsymbol{y'}_1, ..., \boldsymbol{y'}_{m_2'}\}$, we wish to find linear transformations, W_X, W_Y that maximally correlate the subspaces. This can be reduced to an eigenvalue problem. Solving this results in linearly transformed subspaces $\tilde{X}, \tilde{Y}$ with directions $\tilde{\boldsymbol{x}}_i, \tilde{\boldsymbol{y}}_i$ that are maximally correlated with each other, and orthogonal to $\tilde{\boldsymbol{x}}_j, \tilde{\boldsymbol{y}}_j, j < i$. We let $\rho_i = corr(\tilde{\boldsymbol{x}}_i, \tilde{\boldsymbol{y}}_i)$. In summary, we have:

SVCCA Summary

1. Input: X, Y
2. Perform: SVD(X), SVD(Y). Output: $X' = UX, Y' = VY$
3. Perform CCA(X', Y'). Output: $\tilde{X} = W_X X'$, $\tilde{Y} = W_Y Y'$ and $corrs = \{\rho_1, \cdots \rho_{\min(m_1, m_2)}\}$

B Additional Proofs and Figures from Section 2.1

Proof of Orthonormal and Scaling Invariance of CCA:

We can see this using equation (*) as follows: suppose U, V are orthonormal transforms applied to the sets X, Y. Then it follows that Σ_{XX}^a becomes $U\Sigma_{XX}^a U^T$, for $a = \{1, -1, 1/2, -1/2\}$, and similarly for Y and V. Also note Σ_{XY} becomes $U\Sigma_{XY}V^T$. Equation (*) then becomes

$$\tilde{x}_1 = \text{argmax}\left[\frac{x^T U \Sigma_{XX}^{-1/2} \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX} \Sigma_{XX}^{-1/2} U^T x}{||x||}\right]$$

So if $\tilde{u}$ is a solution to equation (*), then $U\tilde{u}$ is a solution to the equation above, which results in the same correlation coefficients.

B.0.1 The importance of SVD: how many directions matter?

While CCA is excellent at identifying useful learned directions that correlate, independent of certain common transforms, it doesn't capture the full picture entirely. Consider the following setting:

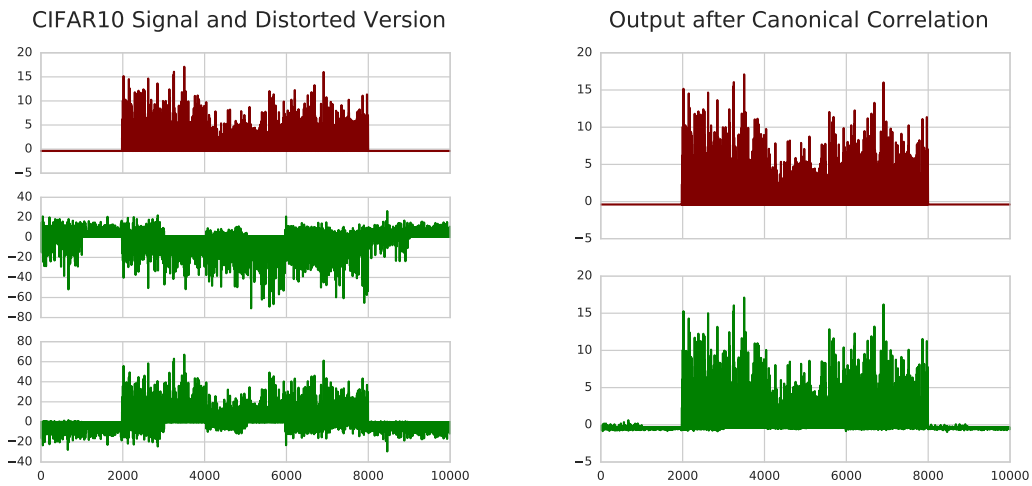

Figure App.1: This figure shows the ability of CCA to deal with orthogonal and scaling transforms. In the first pane, the maroon plot shows one of the highest activation neurons in the penultimate layer of a network trained on CIFAR-10, with the x-axis being (ordered) image ids and the y-axis being activation on that image. The green plots show two resulting distorted directions after this and two of the other top activation neurons are permuted, rotated and scaled. Pane two shows the result of applying CCA to the distorted directions and the original signal, which succeeds in recovering the original signal.

suppose we have subspaces A, B, C, with A being 50 dimensions, B being 200 dimensions, 50 of which are perfectly aligned with A and the other 150 being noise, and C being 200 dimensions, 50 of which are aligned with A (and B) and the other 150 being useful, but different directions.

Then looking at the canonical correlation coefficients of (A, B) and (A, C) will give the *same* result, both being 1 for 50 values and 0 for everything else. But these are two very different cases – the subspace B is indeed well represented by the 50 directions that are aligned with A. But the subspace C has 150 more useful directions.

This distinction becomes particularly important when aggregating canonical correlation coefficients as a measure of similarity, as used in analysing network learning dynamics. However, by first applying SVD to determine the number of directions needed to explain 99% of the observed variance, we can distinguish between pathological cases like the one above.

C Proof of Theorem 1

Here we provide the proofs for Lemma 1, Lemma 2, Theorem 2 and finally Theorem 1.

A preliminary note before we begin:

When we consider a (wlog) n by n channel c of a convolutional layer, we assume it has shape

$$\begin{bmatrix} \boldsymbol{z}_{0,0} & \boldsymbol{z}_{1,2} & \cdots & \boldsymbol{z}_{0,n-1} \\ \boldsymbol{z}_{1,0} & \boldsymbol{z}_{2,2} & \cdots & \boldsymbol{z}_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{z}_{n-1,0} & \boldsymbol{z}_{n-1,1} & \cdots & \boldsymbol{z}_{n-1,n-1} \end{bmatrix}$$